



**COMPETENCY STANDARD**

**FOR**

**INDUSTRIAL ELECTRICAL AND ELECTRONICS**  
**APPLICATION**

**(LIGHT ENGINEERING SECTOR)**

**Skills for Industry Competitiveness and Innovation Program (SICIP)**  
**Finance Division, Ministry of Finance**



## **COPYRIGHT**

The Competency Standards for **Industrial Electrical and Electronics Application** is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as a framework for providing industry-aligned training, ensuring that individuals meeting the prescribed standards through assessments are qualified to secure relevant employment opportunities.

This document has been specifically developed for training purposes under the Skills for Industry Competitiveness and Innovation Program (SICIP), aiming to address the skills requirements of the Light engineering sector. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

While public and private institutions are permitted to use the information from this standard for activities that benefit Bangladesh, other parties must seek prior permission from the document's owner for reproducing its content, either partially or entirely, in English or other languages.

This document is available at:

Skills for Industry Competitiveness and Innovation Program (SICIP)  
Finance Division, Ministry of Finance,  
Probashi Kallyan Bhaban (Level 15 & 16)  
71-72 Eskaton Garden, Dhaka-1000  
Phone: +8802 55138753-55, Fax: +88 02 55138752  
Website: [www.sicip.gov.bd](http://www.sicip.gov.bd)

## INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

**The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.**

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

## OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment, which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

## Units & Elements at a glance:

### Generic Competencies (20 hrs.)

| Code              | Unit of Competency                                                | Elements of Competency                                                                                                                                                                                                         | Duration (Hours) |
|-------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| SICIP-LE-IEE-01-G | Apply Occupational Health and Safety (OHS) Practices in Workplace | <ol style="list-style-type: none"><li>1. Identify OHS policies and procedures</li><li>2. Practice personal health and safety procedures</li><li>3. Report hazards and risks</li><li>4. Respond to emergencies</li></ol>        | 10               |
| SICIP-LE-IEE-02-G | Operate in a Self-Directed Team                                   | <ol style="list-style-type: none"><li>1. Identify team goals and work processes</li><li>2. Communicate and cooperate with team members.</li><li>3. Work as a team member.</li><li>4. Solve problems as a team member</li></ol> | 10               |
| Total Hour        |                                                                   |                                                                                                                                                                                                                                | 20               |

**Sector Specific Competencies (10 hrs.)**

| <b>Code</b>       | <b>Unit of Competency</b> | <b>Elements of Competency</b>                                                                                       | <b>Duration (Hours)</b> |
|-------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------|
| SICIP-LE-IEE-01-S | Apply Green Practices     | 1. Interpret green concepts<br>2. Minimize resource use in the workplace<br>3. Implement waste management practices | 10                      |
| <b>Total Hour</b> |                           |                                                                                                                     | <b>10</b>               |

## Occupation Specific Competencies (270 hrs.)

| Code              | Unit of Competency                                                | Elements of Competency                                                                                                                                                                                                                                                                                                                                               | Duration (hours) |
|-------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| SICIP-LE-IEE-01-O | Implement Basic Electrical Circuits                               | <ol style="list-style-type: none"> <li>1. Interpret electrical symbols, drawings and manual</li> <li>2. Conduct DC circuit experiments</li> <li>3. Construct and implement AC circuits</li> <li>4. Conduct voltage &amp; signal conversion circuit experiments</li> </ol>                                                                                            | 20               |
| SICIP-LE-IEE-02-O | Implement Basic Electronics Circuits                              | <ol style="list-style-type: none"> <li>1. Construct circuits with diodes</li> <li>2. Construct transistor switching circuits</li> <li>3. Conduct experiments with logical circuits</li> <li>4. Perform Op-Amp circuits</li> </ol>                                                                                                                                    | 40               |
| SICIP-LE-IEE-03-O | Execute Electrical Sequence Circuit and Control Using Relay Logic | <ol style="list-style-type: none"> <li>1. Interpret relay logic operation</li> <li>2. Execute timer and counter application</li> <li>3. Construct latch, DOL and star -delta circuits</li> <li>4. Perform automatic forward-reverse operation</li> <li>5. Perform DC motor control operation</li> <li>6. Perform variable frequency drive (VFD) operation</li> </ol> | 45               |
| SICIP-LE-IEE-04-O | Perform PLC Programming and Sequential Control                    | <ol style="list-style-type: none"> <li>1. Carry out PLC wiring</li> <li>2. Setup PLC hardware Configurations with software tool</li> <li>3. Establish communication between PLC and PC</li> <li>4. Prepare ladder diagram for PLC programming</li> <li>5. Perform PLC programming using ladder diagram</li> <li>6. Execute PLC operation</li> </ol>                  | 75               |
| SICIP-LE-IEE-05-O | Design and Implement Microcontroller-based Projects               | <ol style="list-style-type: none"> <li>1. Interpret basic programming for microcontrollers</li> <li>2. Prepare projects using Arduino UNO and NANO.</li> </ol>                                                                                                                                                                                                       | 45               |



| Code               | Unit of Competency                                               | Elements of Competency                                                                                                                                                                                       | Duration (hours) |
|--------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
|                    |                                                                  | 3. Prepare projects using ATmega 2560.<br>4. Prepare temperature measurement project using STM32F411<br>5. Prepare projects using ESP Wroom 32                                                               |                  |
| SICIP-LE-IEE-06-O  | Perform PCB Design using Altium Software                         | 1. Identify Altium interfaces and tools<br>2. Carry out schematic design and diagram for PCB layout<br>3. Perform auto-manual routing and footprint layout<br>4. Design and print PCB projects using Altium. | 30               |
| SICIP-LE-IEE-07-O  | Implement Solar and Wind Energy Systems Using Hybrid Controllers | 1. Interpret solar energy systems<br>2. Design solar home lighting system<br>3. Interpret wind energy systems<br>4. Operate solar & wind energy using hybrid controllers                                     | 15               |
| <b>Total Hours</b> |                                                                  |                                                                                                                                                                                                              | <b>0</b>         |

## COMPETENCY STANDARD: INDUSTRIAL ELECTRICAL AND ELECTRONICS APPLICATION

### The Generic Competencies

|                                                                                                                                                                                                                                                                                                                                                         |                                     |                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE</b>                                                                                                                                                                                                                                              | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-LE-IEE-01-G |
| <b>Unit Descriptor:</b><br>This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in workplace. It specifically includes the tasks of identifying OHS policies and procedures, practicing personal health and safety procedures, reporting hazards and risks, and responding to emergencies. |                                     |                                        |

#### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                            | Performance Criteria                                                                                                                                                                                                                               |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identify OHS policies and procedures           | 1.1 <b><u>OHS policies</u></b> and safe operating procedures are read and understood.<br>1.2 Safety signs and symbols are identified and followed.<br>1.3 Emergency response, evacuation procedures and other contingency measures are determined. |
| 2. Practice personal health and safety procedures | 2.1 OHS policies and procedures are followed and practiced.<br>2.2 <b><u>Personal Protective Equipment (PPE)</u></b> is selected and used.<br>2.3 Personal health, hygiene and safety procedures are practiced.                                    |
| 3. Report hazards and risks                       | 3.1 <b><u>Hazards and risks</u></b> are identified, assessed and controlled.<br>3.2 Incidents arising from hazards and risks are reported to authority.<br>3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.   |
| 4. Respond to emergencies                         | 4.1 Alarms and warning devices are responded.<br>4.2 <b><u>Emergency response plans and procedures</u></b> are implemented.<br>4.3 <b><u>First aid procedure</u></b> is applied during emergency situations.                                       |

#### Range of Variables

| Variable               | Range                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------|
|                        | May include but not limited to:                                                                               |
| 1. OHS policies        | 1.1 International/ Local OHS requirements<br>1.2 Fire Safety Rules and Regulations<br>1.3 Industry Guidelines |
| 2. Personal Protective | 2.1 Apron                                                                                                     |

|                                            |                                                                                                                                                                                  |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Equipment (PPE)                            | 2.2 Gloves<br>2.3 Safety shoes<br>2.4 Helmet<br>2.5 Face mask<br>2.6 Goggles and safety glasses<br>2.7 Ear plugs<br>2.8 Sun block<br>2.9 Chemical/Gas masks                      |
| 3. Hazards and risks                       | 3.1 Chemical hazards.<br>3.2 Biological hazards.<br>3.3 Physical Hazards.<br>3.3.1 Machine hazards.<br>3.3.2 Materials hazards.<br>3.3.3 Tools and Equipment hazards.            |
| 4. Emergency response plans and procedures | 4.1 Firefighting procedures<br>4.2 Earthquake response procedures<br>4.3 Evacuation procedures<br>4.4 Medical and first-aid                                                      |
| 5. First aid procedure                     | 5.1 Washing of open wound<br>5.2 Washing chemically infected area<br>5.3 Applying bandage<br>5.4 Applying CPR (Cardiopulmonary Resuscitation)<br>5.5 Taking appropriate medicine |

### Curricular Content Guide:

|                           |                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 OHS workplace policies and procedures.<br>1.2 Work safety procedures.<br>1.3 Emergency procedures.<br>1.3.1 Firefighting.<br>1.3.2 Earthquake response.<br>1.3.3 Explosion response.<br>1.3.4 Accident response.<br>1.4 Types of (biological, chemical and physical) and their effects.<br>1.5 PPE types and uses.<br>1.6 Personal hygiene practices.<br>1.7 OHS awareness. |
| 2. Underpinning Skills    | 2.1 Identifying OHS policies and procedures<br>2.2 Following personal work safety practices<br>2.3 Reporting hazards and risks<br>2.4 Responding to emergency procedures<br>2.5 Maintaining physical well-being in the workplace<br>2.6 Using firefighting accessories and fire extinguishers<br>2.7 Applying basic first aid procedures                                        |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincerity and honesty to duties                                                                                                                                                                                                                                            |

|                          |                                                                                                                                                                                                                                      |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | 3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Good relationships with peers, sub-ordinates and seniors in the workplace |
| 4. Resource Implications | 4.1 Workplace (simulated or actual)<br>4.2 PPEs<br>4.3 Firefighting equipment<br>4.4 Emergency response manual<br>4.5 First aid kits                                                                                                 |

### Assessment Evidence Guide:

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 followed OHS policies and procedures.<br>1.2 selected and used personal protective equipment (PPE).<br>1.3 practiced personal health and safety procedures.<br>1.4 reported incidents arising from hazards and risks to authority.<br>1.5 implemented plans and procedures to respond emergency.<br>1.6 applied basic first aid procedure. |
| 2. Methods of Assessment          | Methods of assessment may include but not limited to:<br>2.1 Written test<br>2.2 Practical demonstration<br>2.3 Oral question<br>2.4 Portfolio (Optional)                                                                                                                                                                                                                                          |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training<br>3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.                                                                                                                                                  |

|                                                                                                                                                                                                                                                                                                               |                                     |                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>OPERATE IN A SELF-DIRECTED TEAM</b>                                                                                                                                                                                                                                          | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-LE-IEE-02-G |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working and solving problems as a team member. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                          | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identify team goals and work processes       | 1.1 Team goals and collaborative decision-making processes are identified.<br>1.2 Roles and responsibilities of team members are identified.<br>1.3 Relationships within team and with other workers are identified.                                                                                                                                                                                                                                                                 |
| 2. Communicate and cooperate with team members. | 2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives.<br>2.2 Formal and informal <b><u>forms of communication</u></b> are used effectively to support team achievement.<br>2.3 Diversity in character is respected and valued in team functioning.<br>2.4 Views and opinions of other team members are understood, valued and practiced.<br>2.5 Workplace terminology is used correctly to assist communication. |
| 3. Work as a team member.                       | 3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team.<br>3.2 Tasks are performed in accordance with organizational and team requirements, and workplace procedures & practices.<br>3.3 Team member's support to other members is appreciated and valued.<br>3.4 Agreed reporting lines are followed using standard operating procedure.                                                                                |
| 4. Solve problems as a team member              | 4.1 Current and potential problems faced by team are identified.<br>4.2 A solution to the problem is identified.<br>4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.                                                                                                                                                                                                                                                                    |

### Range of Variables

| Variable | Range                           |
|----------|---------------------------------|
|          | May Include but not limited to: |

|                           |                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Forms of communication | <ul style="list-style-type: none"> <li>1.1 Agenda</li> <li>1.2 Simple reports such as progress and incident reports</li> <li>1.3 Job sheets</li> <li>1.4 Operational manuals</li> <li>1.5 Brochures and promotional material</li> <li>1.6 Visual and graphic materials</li> <li>1.7 Standards</li> <li>1.8 OSH information</li> <li>1.9 Signs</li> </ul> |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Curricular Content Guide:

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | <ul style="list-style-type: none"> <li>1.1 Team goals and collaborative decision-making processes</li> <li>1.2 Roles and responsibilities of team members</li> <li>1.3 Relationships within team and with other workers</li> <li>1.4 Effective interpersonal skills to interact with team members</li> <li>1.5 Effective formal and informal forms of communication</li> <li>1.6 Value of diversity in team functioning.</li> <li>1.7 Correct use of workplace terminology</li> <li>1.8 Team's duties, responsibilities, authorities, objectives and task requirements</li> <li>1.9 Support mechanism to other members of team to ensure achievements of goals.</li> <li>1.10 Methods of identifying current and potential problems faced by a team</li> <li>1.11 Effective problem-solving methods and evaluation of outcomes</li> </ul>                                                                                                            |
| 2. Underpinning Skills    | <ul style="list-style-type: none"> <li>2.1 Identifying team goals and collaborative decision-making processes</li> <li>2.2 Identifying roles and responsibilities of team members</li> <li>2.3 Identifying relationships within team and with other workers</li> <li>2.4 Using effective interpersonal skills to interact with team members and to contribute to activities and objectives</li> <li>2.5 Using formal and informal forms of communication</li> <li>2.6 Understanding and valuing views and opinions of other team members</li> <li>2.7 Performing tasks in accordance with organizational and team requirements, and workplace procedures and practice.</li> <li>2.8 Team member's support to other members is appreciated and valued Identifying current and potential problems faced by the team</li> <li>2.9 Identifying solutions to the problem</li> <li>2.10 Solving problems effectively and evaluating the outcome</li> </ul> |

|                           |                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | of the implemented solution                                                                                                                                                                                                                                                                                                                                                  |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincerity and honesty to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Good relationships with peers, sub-ordinates and seniors in the workplace |
| 4. Resource Implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)<br>4.2 Pens<br>4.3 Papers<br>4.4 Work books<br>4.5 Learning manuals                                                                                                                                                                                                                         |

#### **Assessment Evidence Guide:**

|                                   |                                                                                                                                                                                                                                                  |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 Identified team goals and work processes<br>1.2 Communicated and cooperated with team members.<br>1.3 Worked as a team member<br>1.4 Solved problems as a team member                    |
| 2. Methods of Assessment          | Methods of assessment may include but not limited to:<br>2.1 Written test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                                                                                     |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training<br>3.2 Assessment should be done by a certified assessor or occupation-specific industry expert |

## The Sector Specific Competencies

|                                                                                                                                                                                                                                                                              |                                     |                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>APPLY GREEN PRACTICES</b>                                                                                                                                                                                                                   | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-LE-IEE-01-S |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace, and implementing waste management practices. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                    | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Interpret green concepts               | 1.1 <b><u>Principles of green practices</u></b> are explained.<br>1.2 <b><u>Sources of environmental impacts</u></b> during light engineering activities are identified.<br>1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained.<br>1.4 <b><u>Ways of mitigating environmental impacts</u></b> in light engineering sector are explained. |
| 2. Minimize resource use in the workplace | 2.1 Water, energy, and raw material consumption are documented.<br>2.2 <b><u>Recyclable and non-recyclable items</u></b> are identified.<br>2.3 Procedures to reduce resource consumption are implemented.<br>2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.                                                                                                                      |
| 3. Implement waste management practices   | 3.1 <b><u>Different types of wastes</u></b> are identified.<br>3.2 Hazardous waste is disposed of according to environmental regulations.<br>3.3 Green habits to reduce wastes in personal and professional life are practiced.                                                                                                                                                                                            |

### Range of Variables

| Variable                         | Range                                                                                                                                                                                                                             |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | May include but not limited to:                                                                                                                                                                                                   |
| 1. Principles of green practices | 1.1 Reducing energy consumption<br>1.2 6R for waste management<br>1.2.1 Refuse<br>1.2.2 Reduce<br>1.2.3 Reuse<br>1.2.4 Recycle<br>1.2.5 Recover<br>1.2.6 Repair<br>1.3 use of sustainable materials with low environmental impact |



|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | 1.4 Recycling materials<br>1.5 Sustainable transportation                                                                                                                                                                                                                                                                                                                                                                                 |
| 2. Sources of environmental impacts         | 2.1 Air pollution<br>2.1.1 Dust generation<br>2.1.2 Emission from machinery and vehicles<br>2.2 Water pollution<br>2.2.1 Debris and sediments<br>2.2.2 Chemical's leakage<br>2.3 Noise pollution<br>2.4 Waste generation<br>2.4.1 waste<br>2.4.2 Non-recyclable and hazardous materials<br>2.5 Resource Depletion<br>2.5.1 Excessive use of raw materials<br>2.5.2 Non-renewable material uses like use of fossil fuel, steel cement etc. |
| 3. Ways of mitigating environmental impacts | 3.1 Utilizing Energy-Efficient Equipment<br>3.2 Adopting Renewable Energy Sources<br>3.3 Implementing Site Protection Measures<br>3.4 Using Reusable Materials<br>3.5 Choosing Sustainable Materials<br>3.6 Using Noise-Reducing Equipment and scheduling work appropriately<br>3.7 Optimizing Logistics and Delivery                                                                                                                     |
| 4. Recyclable and non-recyclable items      | 4.1 Recyclable items<br>4.1.1 Metal<br>4.1.2 Wood<br>4.1.3 Brick<br>4.1.4 Concrete<br>4.2 Non-recyclable items<br>4.2.1 Paints and coatings<br>4.2.2 Contaminated materials like lead paint, asbestos<br>4.2.3 Treated wood                                                                                                                                                                                                               |
| 5. Different types of waste                 | 5.1 Wood waste<br>5.2 Metal waste<br>5.3 Plastic waste<br>5.4 Electrical system waste<br>5.5 Gypsum board waste<br>5.6 Packaging waste<br>5.7 Organic waste<br>5.8 Chemical West<br>5.9 Fuel/Oil west                                                                                                                                                                                                                                     |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Principles of green practices in Light Engineering sector<br>1.2 Key terms and symbols related to environmental sustainability in Light Engineering drawings<br>1.3 Sources of environmental impacts in Light Engineering |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | 1.4 Methods to minimize resource consumption (water, energy, raw materials)<br>1.5 Waste management practices<br>1.6 Sustainable alternatives to fossil-based energy resources<br>1.7 Personal and workplace habits for reducing environmental impact                                                                                                                                                             |
| 2. Underpinning Skills    | 2.1 Identifying environmental impacts in Light Engineering activities<br>2.2 Applying methods to minimize resource use in the workplace<br>2.3 Implementing waste management practices<br>2.4 Following procedures to safely dispose of hazardous materials<br>2.5 Documenting water, energy, and material consumption in the workplace<br>2.6 Performing green practices in personal and professional activities |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincerity and honesty to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Good relationships with peers, sub-ordinates and seniors in the workplace                                      |
| 4. Resource Implications  | 4.1 Workplace (simulated or actual)<br>4.2 Different types of Light Engineering hand tools and power tools<br>4.3 Safety gear<br>4.4 Pens<br>4.5 Papers<br>4.6 Work books<br>4.7 Operation and maintenance manuals<br>4.8 Waste segregation bins                                                                                                                                                                  |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 Explained the principles of green practices in Light Engineering<br>1.2 Identified sources of environmental impacts<br>1.3 Implemented procedures to minimize resource use (water, energy, raw materials)<br>1.4 Disposed of hazardous waste in compliance with safety and environmental standards |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                          |                                                                                                                                                                                                                                                               |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | 1.5 Practiced green habits to reduce waste in both personal and professional life                                                                                                                                                                             |
| 2. Methods of Assessment | Competency should be assessed by:<br>2.1 Written examination<br>2.2 Demonstration<br>2.3 Oral questioning<br>2.4 Workplace observation                                                                                                                        |
| 3. Context of Assessment | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert. |

## Occupation-specific Competencies

|                                                                                                                                                                                                                                                                                                                                                                                |                                     |                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>IMPLEMENT BASIC ELECTRICAL CIRCUITS</b>                                                                                                                                                                                                                                                                                                       | <b>Nominal Duration:</b><br>20 hrs. | <b>Unit Code:</b><br>SICIP-LE-IEE-01-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to implement basic electrical circuits. It specifically includes the tasks of interpreting electrical symbols, drawings and manuals, conducting DC circuit experiments, constructing and implementing AC circuits and conducting voltage & signal conversion circuit experiments. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

| Elements of Competency                                     | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Interpret electrical symbols, drawings and manuals      | 1.1 <b><u>Electrical symbols</u></b> used in circuit diagrams are identified and understood.<br>1.2 <b><u>Electrical drawings</u></b> are identified to implement wiring and troubleshooting procedures.<br>1.3 Electrical manuals are identified and understood.<br>1.4 Wiring diagrams and schematics are understood.<br>1.5 Common electrical standards and codes are recognized.                                                                                                   |
| 2. Conduct DC circuit experiments                          | 2.1 Ohm's Law is verified by measuring voltage, current, and resistance in a DC circuit.<br>2.2 Series circuits are constructed and tested.<br>2.3 Parallel circuits are constructed and tested.<br>2.4 Combined circuits (series and parallel) are created.<br>2.5 <b><u>Circuit theorems</u></b> are implemented and verified.                                                                                                                                                       |
| 3. Construct and implement AC circuits                     | 3.1 RL circuits are constructed and implemented.<br>3.2 RC circuits are constructed and implemented.<br>3.3 Resistor (R) and capacitor (C) components are connected.<br>3.4 RLC circuits are constructed and implemented.<br>3.5 <b><u>Resonance circuits</u></b> are designed and implemented.<br>3.6 <b><u>Filter circuits</u></b> are constructed and tested.                                                                                                                       |
| 4. Conduct voltage & signal conversion circuit experiments | 4.1 A/D (Analog to Digital) converter circuit is constructed and tested.<br>4.2 D/A (Digital to Analog) converter circuit is constructed and tested.<br>4.3 Voltage conversion is conducted using a <b><u>transformer</u></b> .<br>4.4 <b><u>Conversion characteristics</u></b> are measured and analyzed.<br>4.5 Signal integrity is verified by ensuring minimal distortion, noise, or loss during the conversion process.<br>4.6 Impact of load variation on conversion accuracy is |

|  |           |
|--|-----------|
|  | assessed. |
|--|-----------|

### Range of Variables

| Variable                      | Range (Includes but not limited to:)                                                                                                                                                          |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Electrical symbols         | 1.1 Source (AC/DC)<br>1.2 Resistor<br>1.3 Capacitor<br>1.4 Inductor<br>1.5 Switches<br>1.6 Relay coils<br>1.7 Magnetic contactor<br>1.8 Circuit breaker<br>1.9 Timer/counter<br>1.10 Overload |
| 2. Electrical drawings        | 2.1 Layout diagram<br>2.2 Schematic diagram<br>2.3 Wiring diagram                                                                                                                             |
| 3. Circuit theorems           | 3.1 Kirchhoff's voltage law<br>3.2 Kirchhoff's current law<br>3.3 Thevenin's theorem<br>3.4 Superposition                                                                                     |
| 4. Resonance circuits         | 4.1 Series resonance<br>4.2 Parallel resonance                                                                                                                                                |
| 5. Filter circuits            | 5.1 Active filter<br>5.2 Passive filter                                                                                                                                                       |
| 6. Transformer                | 6.1 Step-up<br>6.2 Step-down                                                                                                                                                                  |
| 7. Conversion characteristics | 7.1 Sampling rate<br>7.2 Resolution for A/D<br>7.3 D/A converters                                                                                                                             |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Identify standard electrical symbols used in circuit diagrams and manuals for installation and diagnostics.<br>1.2 Understand the function of electrical schematics and wiring diagrams for interpreting system layouts.<br>1.3 Define Ohm's Law and its practical application through voltage, current, and resistance relationships.<br>1.4 Explain the characteristics and behavior of series, parallel, and combined electrical circuits.<br>1.5 Describe the theoretical principles of RL, RC, and RLC circuits and their impact on AC/DC systems.<br>1.6 Understand the concept of electrical resonance and its implementation in tuned circuits.<br>1.7 Explain the structure and function of analog-to-digital (A/D) and digital-to-analog (D/A) converter circuits. |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | 1.8 Identify the role and working principle of transformers in voltage conversion applications.<br>1.9 Describe the effects of load variation and signal distortion.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2. Underpinning Skills    | 2.1 Applying Ohm's Law by measuring voltage, current, and resistance in DC circuits.<br>2.2 Constructing series, parallel, and combination circuits and verifying their functionality.<br>2.3 Implementing basic circuit theorems in practical electrical experiments.<br>2.4 Constructing and testing RL, RC, and RLC circuits using standard components.<br>2.5 Designing and implementing resonance and filter circuits for signal control.<br>2.6 Constructing and verifying A/D and D/A converter circuits for signal processing.<br>2.7 Conducting voltage conversion using transformers and analyzing conversion characteristics.<br>2.8 Assessing signal integrity and evaluating the effect of load variation on circuit performance. |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincerity and honesty to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Good relationships with peers, sub-ordinates and seniors in the workplace                                                                                                                                                                                                                                                                                                                                                                   |
| 4. Resource Implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)<br>4.2 Tools, equipment and facilities appropriate to the process or activities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                     |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 interpreted electrical symbols, drawings and manual<br>1.2 conducted DC circuit experiments<br>1.3 constructed and implemented AC circuits<br>1.4 conducted voltage & signal conversion circuit experiments |
| 2. Methods of Assessment          | Competency should be assessed by:<br>2.1 Written Test<br>2.2 Practical demonstration<br>2.3 Oral questioning                                                                                                                                                        |

|                          |                                                                                                                                                                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | 2.4 Portfolio (Optional)                                                                                                                                                                                                                                                 |
| 3. Context of Assessment | <p>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</p> <p>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</p> |

|                                                                                                                                                                                                                                                                                                                                          |                                     |                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>IMPLEMENT BASIC ELECTRONICS CIRCUITS</b>                                                                                                                                                                                                                                                                | <b>Nominal Duration:</b><br>40 Hrs. | <b>Unit Code:</b><br>SICIP-LE-IEE-02-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to implement basic electronics circuits. It specifically includes the tasks of constructing circuits with diodes, constructing transistor switching circuits, conducting experiments with logical circuits, and performing Op-Amp circuits. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

| Elements of Competency                       | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Construct circuits with diodes            | 1.1 <b><u>Semiconductor devices</u></b> are identified and explained.<br>1.2 Function of P-N junction diode are explained.<br>1.3 <b><u>Half-wave rectifier circuit</u></b> is constructed and observed the I/O wave form.<br>1.4 <b><u>Full-wave rectifier circuit</u></b> is constructed and observed the I/O wave form.<br>1.5 Zener diode circuits are constructed for voltage regulation.<br>1.6 Shockley diode circuits are constructed and observed.<br>1.7 Output of the rectifiers is evaluated.                                                                                                                                                                                                                |
| 2. Construct transistor switching circuits   | 2.1 Working principle of transistors (BJT and FET) is understood.<br>2.2 <b><u>Field Effect Transistor (FET)</u></b> circuits are implemented to control current using voltage (gate-source voltage).<br>2.3 Silicon Controlled Rectifier (SCR) circuits are implemented, ensuring proper triggering.<br>2.4 Diodes for Alternating Current (DIAC) circuits are constructed with appropriate voltage and current handling characteristics.<br>2.5 Triode for Alternating Current (TRIAC) circuits are implemented for AC power control,<br>2.6 Unijunction Transistor (UJT) circuits are implemented.<br>2.7 <b><u>Switching characteristic devices</u></b> are understood and verified to use with appropriate circuit. |
| 3. Conduct experiments with logical circuits | 3.1 <b><u>Logic gates</u></b> operations are understood.<br>3.2 Half adder circuit is constructed and tested.<br>3.3 Full adder circuit is constructed and tested.<br>3.4 Multiplexer circuits are constructed ensuring correct input based on the selection lines.<br>3.5 De-multiplexer circuits are constructed and verified for                                                                                                                                                                                                                                                                                                                                                                                      |



|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>different input conditions.</p> <p>3.6 Truth tables for each circuit are created and verified.</p> <p>3.7 Any errors in connections or logic are identified and corrected to achieve the desired results.</p> <p>3.8 Practical applications of logical circuits are understood.</p>                                                                                                                                                                                                                                                                                                          |
| 4. Perform Op-Amp circuits | <p>4.1 Inverting amplifier and non-inverting amplifier circuits are designed and tested with correct feedback and input resistor configurations.</p> <p>4.2 Differentiator and Integrator amplifier circuits are built and tested.</p> <p>4.3 Adder and subtractor circuits are constructed using op-amps (Operational Amplifier).</p> <p>4.4 <b>Filter circuits</b> are designed and tested.</p> <p>4.5 Performance of each circuit is verified by applying test signals and measuring the output.</p> <p>4.6 Troubleshooting of op-amp circuits is performed, ensuring correct operation.</p> |

#### Range of Variables

| Variable                           | Range (Includes but not limited to:)                                                                                                                                                                                                            |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Semiconductor devices           | <p>1.1 Diodes</p> <p>1.2 Transistors</p> <p>1.3 Field Effect Transistor (FET)</p> <p>1.4 Diodes for Alternating Current (DIAC)</p> <p>1.5 Triode for Alternating Current (TRIAC)</p> <p>1.6 Silicon Controlled Rectifier (SCR) or Thyristor</p> |
| 2. Half-wave rectifier circuit     | <p>2.1 Center tapped circuit using one diode</p> <p>2.2 Rectifier with single diode</p>                                                                                                                                                         |
| 3. Full-wave rectifier circuit     | <p>3.1 Center tapped</p> <p>3.2 Bridge</p>                                                                                                                                                                                                      |
| 4. Field Effect Transistor (FET)   | <p>4.1 Junction Field Effect Transistor (JFET)</p> <p>4.2 Metal Oxide Semiconductor Field Effect Transistor (MOSFET)</p>                                                                                                                        |
| 5. Switching characteristic device | <p>5.1 Transistor</p> <p>5.2 FET</p> <p>5.3 SCR</p> <p>5.4 DIAC</p> <p>5.5 TRIAC</p> <p>5.6 UJT</p>                                                                                                                                             |
| 6. Logic gates                     | <p>6.1 AND</p> <p>6.2 OR</p> <p>6.3 NOT</p> <p>6.4 XNOR</p> <p>6.5 XOR</p> <p>6.6 NAND</p>                                                                                                                                                      |
| 7. Filter circuits                 | <p>7.1 Low-pass</p> <p>7.2 High-pass</p>                                                                                                                                                                                                        |

|  |               |
|--|---------------|
|  | 7.3 Band-pass |
|  | 7.4 Band-stop |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Identify various semiconductor devices.<br>1.2 Understand the concept and behavior of P-N junctions and their application in rectifier and regulator circuits.<br>1.3 Define the structure and working principles of half-wave and full-wave rectifiers through waveform analysis.<br>1.4 Explain voltage regulation using Zener diodes and the triggering process in SCR circuits.<br>1.5 Describe the operation of transistor-based circuits for current control and signal amplification.<br>1.6 Understand logic gates, their truth tables, and their applications.<br>1.7 Procedure for constructing and verifying adder, subtractor.<br>1.8 Process of troubleshooting analog and digital circuits.                                                                         |
| 2. Underpinning Skills    | 2.1 Constructing half-wave and full-wave rectifier circuits and observing their input/output waveforms.<br>2.2 Implementing Zener, Shockley, SCR, DIAC, TRIAC, and UJT circuits for practical power control tasks.<br>2.3 Applying the working principle of transistors (BJT, FET) in current/voltage-controlled operations.<br>2.4 Constructing logic gates, half adder, and full adder circuits and verifying their truth tables.<br>2.5 Building multiplexer and de-multiplexer circuits and ensuring correct switching logic.<br>2.6 Designing and testing op-amp-based circuits.<br>2.7 Implementing and verifying filter circuits for signal processing applications.<br>2.8 Conducting performance testing and performing troubleshooting to ensure correct circuit operation. |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincerity and honesty to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Good relationships with peers, subordinates and seniors in the workplace                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4. Resource Implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)<br>4.2 Tools, equipment, materials and facilities appropriate to the process or activities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                               |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br><br>1.1 constructed circuits with diodes<br>1.2 constructed transistor switching circuits<br>1.3 conducted experiments with logical circuits<br>1.4 performed Op-Amp circuits                             |
| 2 Methods of Assessment           | Competency should be assessed by:<br><br>2.1 Written test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                                                                                                                  |
| 3 Context of Assessment           | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert. |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                         |                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>EXECUTE ELECTRICAL SEQUENCE<br/>CIRCUIT AND CONTROL USING RELAY<br/>LOGIC</b>                                                                                                                                                                                                                                                                                                                                                                          | <b>Nominal<br/>Duration:</b><br>45 hrs. | <b>Unit Code:</b><br>SICIP-LE-IEE-03-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to execute electrical sequence circuit and control using relay logic. It specifically includes the tasks of interpreting relay logic operation, executing timer and counter application, constructing latch, DOL and star -delta circuits, performing automatic forward- reverse operation, performing DC motor control operation and performing variable frequency drive (VFD) operation. |                                         |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

| Elements of Competency                           | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Interpret relay logic operation               | 1.1 Fundamental concepts of relay logic are identified and explained.<br>1.2 <b><u>Components of industrial switching devices</u></b> are identified.<br>1.3 Functions of Industrial switching devices are described.<br>1.4 Operation of relay logic circuits is interpreted.<br>1.5 Common relay <b><u>logic symbols</u></b> and notations are identified.<br>1.6 Application of relay logic in industrial automation processes is described.                                                                                   |
| 2. Execute timer and counter application         | 2.1 <b><u>Timer</u></b> and <b><u>counter</u></b> applications are selected based on control system needs.<br>2.2 Timer and counter settings are configured according to application requirements.<br>2.3 Timer and counter operations are executed following standard procedures.<br>2.4 Timer and counter outputs are monitored and verified.<br>2.5 Faults or deviations in timer and counter performance are identified.                                                                                                      |
| 3. Construct latch, DOL and star -delta circuits | 3.1 Circuit diagrams for latch, DOL, and star-delta configurations are interpreted.<br>3.2 Latch circuits are constructed by connecting relays, contacts, and coils as per standard wiring diagrams.<br>3.3 DOL starter circuits are assembled ensuring proper connections for motor start and stop functions.<br>3.4 Star-delta starter circuits are constructed following established wiring procedures.<br>3.5 Circuits are tested for correct operation.<br>3.6 Induction motor is operated using DOL and star-delta starter. |

|                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4. Perform automatic forward- reverse operation     | <p>4.1 Control circuit for forward and reverse operation is interpreted.</p> <p>4.2 Automatic forward and reverse operation are executed using timer and counter.</p> <p>4.3 Operation is tested to verify smooth and correct transition between forward and reverse motion.</p> <p>4.4 Induction motor is operated using automatic forward-reverse starter.</p> <p>4.5 Faults or abnormalities during operation are identified and reported for corrective actions.</p>                                                |
| 5. Perform DC motor control operation               | <p>5.1 DC series motor is configured and operated with correct connection of armature and field windings.</p> <p>5.2 DC shunt motor is configured and operated with proper connection of armature and field windings.</p> <p>5.3 DC compound motor is configured and operated with correct configuration of both series.</p> <p>5.4 <b><u>Motor operation parameters</u></b> are monitored and adjusted.</p> <p>5.5 <b><u>Control methods</u></b> are applied.</p> <p>5.6 Motor basic troubleshooting is performed.</p> |
| 6. Perform variable frequency drive (VFD) operation | <p>6.1 VFD wiring is performed according to the manufacturer's specifications.</p> <p>6.2 Parameters of VFD are set according to the operating requirements.</p> <p>6.3 Induction motor is operated at different speeds, ensuring the VFD correctly adjusts motor speed.</p> <p>6.4 Induction motor is operated with an analog signal, ensuring proper motor speed control.</p> <p>6.5 Induction motor is operated in multi-speed mode, ensuring the motor runs at various predefined speeds.</p>                       |

### Range of Variables

| Variable                                      | Range (Includes but not limited to:)                                                                                                                      |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Components of industrial switching devices | <p>1.1 Switches</p> <p>1.2 Relay</p> <p>1.3 Magnetic Contactor</p> <p>1.4 Timer</p> <p>1.5 Counter</p> <p>1.6 LED</p> <p>1.7 Motor</p> <p>1.8 Sensors</p> |
| 2. Logic symbols                              | <p>2.1 Normally Open (NO)</p> <p>2.2 Normally Close (NC)</p> <p>2.3 Timer</p> <p>2.4 Counter</p> <p>2.5 Relay Coil</p>                                    |

|                               |                                                                                            |
|-------------------------------|--------------------------------------------------------------------------------------------|
| 3. Timer                      | 3.1 On-delay timer<br>3.2 Off-delay timer                                                  |
| 4. Counter                    | 4.1 Up counter<br>4.2 Down counter                                                         |
| 5. Motor operation parameters | 5.1 Voltage<br>5.2 Current<br>5.3 Resistance<br>5.4 Rated power<br>5.5 Speed<br>5.6 Torque |
| 6. Control methods            | 6.1 Voltage<br>6.2 Field flux control<br>6.3 Resistance                                    |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Identify the fundamental concepts and working principles of relay logic systems.<br>1.2 Describe the functions and configurations of industrial switching devices.<br>1.3 Explain the standard symbols and notations used in relay logic circuits.<br>1.4 Understand the applications of relay logic in industrial automation processes.<br>1.5 Define the role of timers and counters in controlling sequential operations.<br>1.6 Conceptualize the wiring and operation of control circuits like latch, DOL, and star-delta starters.<br>1.7 Identify the procedure for forward-reverse control using timers and counters.<br>1.8 Explain the operational characteristics of induction and DC motors with various starters.<br>1.9 Understand the purpose and function of VFDs in controlling motor speed and operation modes. |
| 2. Underpinning Skills    | 2.1 Constructing latch, DOL, and star-delta circuits following standard wiring diagrams.<br>2.2 Configuring timer and counter settings and executing their operations as per control system needs.<br>2.3 Testing circuits for correct operation and monitoring output performance of timers and counters.<br>2.4 Operating induction motors using DOL, star-delta, and automatic forward-reverse starters.<br>2.5 Implementing control circuits for forward and reverse motor operations using timers and counters.<br>2.6 Configuring and operating DC series, shunt, and compound motors with proper armature-field connections.<br>2.7 Performing VFD wiring and setting parameters for                                                                                                                                           |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>accurate speed control and analog signal response.</p> <p>2.8 Applying basic troubleshooting techniques and monitoring motor performance under various conditions.</p>                                                                                                                                                                                                                               |
| 3. Underpinning Attitudes | <p>3.1 Commitment to occupational health and safety</p> <p>3.2 Promptness in carrying out activities</p> <p>3.3 Sincerity and honesty to duties</p> <p>3.4 Environmental concerns</p> <p>3.5 Eagerness to learn</p> <p>3.6 Tidiness and timeliness</p> <p>3.7 Respect for rights of peers and seniors in workplace</p> <p>3.8 Good relationships with peers, sub-ordinates and seniors in workplace</p> |
| 4 Resource Implications   | <p>The following resources must be provided:</p> <p>4.1 Workplace (simulated or actual)</p> <p>4.2 Tools and equipment required for the job</p> <p>4.3 Materials required</p>                                                                                                                                                                                                                           |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | <p>Assessment required evidence that the candidate:</p> <p>1.1 interpreted relay logic operation</p> <p>1.2 executed timer and counter application</p> <p>1.3 constructed latch, DOL and star -delta circuits</p> <p>1.4 performed automatic forward- reverse operation</p> <p>1.5 performed DC motor control operation</p> <p>1.6 performed variable frequency drive (VFD) operation</p> |
| 2. Methods of Assessment          | <p>Competency should be assessed by:</p> <p>2.1 Written Test</p> <p>2.2 Practical demonstration</p> <p>2.3 Oral questioning</p> <p>2.4 Portfolio (Optional)</p>                                                                                                                                                                                                                           |
| 3. Context of Assessment          | <p>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</p> <p>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</p>                                                                                                                  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                     |                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>PERFORM PLC PROGRAMMING<br/>AND SEQUENTIAL CONTROL</b>                                                                                                                                                                                                                                                                                                                                                             | <b>Nominal Duration:</b><br>75 hrs. | <b>Unit Code:</b><br>SICIP-LE-IEE-04-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to perform PLC programming and sequential control. It specifically includes the tasks of carrying out PLC wiring, setting up PLC hardware configurations with software tool, establishing communication between PLC and PC, preparing ladder diagram for PLC programming, performing PLC programming using ladder diagram and executing PLC operation. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

| Elements of Competency                                  | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Carry out PLC wiring                                 | 1.1 PLC wiring configuration is interpreted.<br>1.2 Sinking and sourcing for PLC inputs/outputs are identified.<br>1.3 Biasing power connections to the PLC are performed, ensuring the PLC receives the correct voltage.<br>1.4 Input power connections are performed, ensuring the power supply is properly connected.<br>1.5 Output power connections are performed, ensuring the power supply is correctly connected.<br>1.6 Wiring is tested for continuity and correct operation, ensuring all connections are secure and functional.                                                                                     |
| 2. Setup PLC hardware Configurations with software tool | 2.1 Typical PLC components and the control devices are installed as per job requirement.<br>2.2 Sensors and switches are connected to a particular input and output of PLC.<br>2.3 <b><u>Cable connections</u></b> between the PLC and the <b><u>programming devices</u></b> are connected as per work instruction.<br>2.4 Connection is checked as per required specification.<br>2.5 <b><u>Protective devices</u></b> are set as per the user manual.<br>2.6 Required <b><u>software tool</u></b> is installed.<br>2.7 Address of PLC and PC are configured.<br>2.8 Selected hardware is configured using PLC software tools. |
| 3. Establish communication between PLC and PC           | 3.1 Hardware devices are checked in accordance with user manual.<br>3.2 Communication between PLC and PC using <b><u>communication protocols</u></b> is established.<br>3.3 Communication between PLC & PC is verified.                                                                                                                                                                                                                                                                                                                                                                                                         |
| 4. Prepare ladder diagram for PLC programming           | 4.1 Necessary <b><u>software and hardware</u></b> are selected as per job requirements.<br>4.2 Data type, instruction, memory, special memory,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>comparators and math function are described.</p> <p>4.3 Table chart for sub-program, function, function block and data block are prepared.</p> <p>4.4 <b><u>Programming components and ladder logic symbols</u></b> are identified required for ladder diagram of PLC.</p>                                                                                                                                                                                                                                                   |
| 5. Perform PLC programming using ladder diagram | <p>5.1 Basic ladder logic symbols and components are identified according to instruction manual.</p> <p>5.2 Ladder diagram is constructed by arranging symbols logically to represent the desired control sequence.</p> <p>5.3 <b><u>Input and output devices</u></b> are tested.</p> <p>5.4 Input and output devices are mapped within the ladder diagram.</p> <p>5.5 Ladder diagram is checked for logical consistency.</p> <p>5.6 Modifications and corrections are made to the ladder diagram based on testing results.</p> |
| 6. Execute PLC operation                        | <p>6.1 PLC program is loaded onto the controller, ensuring the correct logic and operations.</p> <p>6.2 Inputs are monitored and the corresponding outputs are activated.</p> <p>6.3 Real-time control is implemented, ensuring that the PLC responds.</p> <p>6.4 PLC diagnostic system options are identified and used to monitor system performance.</p> <p>6.5 Performance is verified by checking the operation of the PLC in controlling.</p> <p>6.6 Program modifications are made, if necessary.</p>                     |

### Range of Variables

| Variable                   | Range (Includes but not limited to:)                                                      |
|----------------------------|-------------------------------------------------------------------------------------------|
| 1. Cable connections       | <p>1.1 RS 232</p> <p>1.2 USB</p> <p>1.3 Ethernet</p> <p>1.4 RS485</p> <p>1.5 Profinet</p> |
| 2. Programming devices     | <p>2.1 Computer with programming software</p> <p>2.2 Laptop with programming software</p> |
| 3. Protective devices      | <p>3.1 Fuse</p> <p>3.2 Miniature Circuit Breaker (MCB)</p>                                |
| 4. Software tool           | <p>4.1 TIA Portal</p> <p>4.2 GX Works</p> <p>4.3 WPLSoft</p>                              |
| 5. Communication protocols | <p>5.1 Profinet</p> <p>5.2 USB</p> <p>5.3 Modbus</p>                                      |
| 6. Software and hardware   | <p>6.1 <b>Software</b></p> <p>6.1.1 TIA Portal</p>                                        |

|                                                    |                                                                                                                                                                                                                                               |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | 6.1.2 GX Works<br>6.1.3 WPLSoft<br><b>6.2 Hardware</b><br>6.2.1 PLC CPU<br>6.2.2 Input module<br>6.2.3 Output module<br>6.2.4 Communication module<br>6.2.5 Communication cable<br>6.2.6 Programming cable                                    |
| 7. Programming components and ladder logic symbols | 7.1 Make contact<br>7.2 Break contact<br>7.3 Analog contact<br>7.4 Relay coil<br>7.5 Inverted output<br>7.6 Analog output<br>7.7 All types of timers<br>7.8 Counter<br>7.9 Comparator<br>7.10 Mathematical functions<br>7.11 Data converter   |
| 8. Input and output devices                        | <b>8.1 Input</b><br>8.1.1 Push button<br>8.1.2 Proximity sensor<br>8.1.3 Photo sensor<br>8.1.4 Analog sensor<br><b>8.2 Output</b><br>8.2.1 Relay<br>8.2.2 Magnetic contactor<br>8.2.3 Variable Frequency Drive (VFD)<br>8.2.4 Induction motor |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Identify PLC input/output types including sinking and sourcing methods.<br>1.2 Describe the components and wiring configurations used in PLC systems.<br>1.3 Explain the process of connecting sensors, switches, and power supplies to PLC hardware.<br>1.4 Define the use of communication protocols for PLC and PC interaction.<br>1.5 Understand how to install and configure PLC software and address settings.<br>1.6 Identify ladder logic symbols, sub-programs, and functional blocks used in PLC programming.<br>1.7 Describe how inputs and outputs are mapped and tested within a ladder diagram.<br>1.8 Conceptualize the steps for constructing, checking, and |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>modifying a ladder diagram.</p> <p>1.9 Explain how PLC diagnostics and real-time control functions help verify system performance.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2. Underpinning Skills    | <p>2.1 Performing PLC wiring, power connections, and continuity tests to ensure correct voltage and operation.</p> <p>2.2 Installing typical PLC components, sensors, switches, and control devices as per job requirements.</p> <p>2.3 Configuring addresses, protective devices, and hardware/software tools for PLC setup.</p> <p>2.4 Establishing and verifying communication between PLC and PC using standard protocols.</p> <p>2.5 Constructing ladder diagrams using logic symbols to represent desired control sequences.</p> <p>2.6 Mapping input/output devices within the ladder program and testing them for functionality.</p> <p>2.7 Implementing PLC programs on the controller and monitoring real-time input-output behavior.</p> <p>2.8 Applying diagnostic tools to evaluate PLC performance and troubleshoot operational issues.</p> <p>2.9 Modifying and updating ladder diagrams and programs based on test results and performance feedback.</p> |
| 3. Underpinning Attitudes | <p>3.1 Commitment to occupational health and safety</p> <p>3.2 Promptness in carrying out activities</p> <p>3.3 Sincerity and honesty to duties</p> <p>3.4 Environmental concerns</p> <p>3.5 Eagerness to learn</p> <p>3.6 Tidiness and timeliness</p> <p>3.7 Respect for rights of peers and seniors in workplace</p> <p>3.8 Good relationships with peers, sub ordinates and seniors in workplace</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 4. Resource Implications  | <p>The following resources must be provided:</p> <p>4.1 Workplace (simulated or actual)</p> <p>4.2 Tools, equipment and facilities needed for the process/ activity</p> <p>4.3 Materials required</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | <p>Assessment required evidence that the candidate:</p> <p>1.1 carried out PLC wiring</p> <p>1.2 setup PLC hardware Configurations with software tool</p> <p>1.3 established communication between PLC and PC</p> <p>1.4 prepared ladder diagram for PLC programming</p> <p>1.5 performed PLC programming using ladder diagram</p> <p>1.6 executed PLC operation</p> |
| 2. Methods of                     | Competency should be assessed by:                                                                                                                                                                                                                                                                                                                                    |

|                          |                                                                                                                                                                                                                                                               |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Assessment               | 2.1 Written test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                                                                                                                                                           |
| 3. Context of Assessment | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert. |

|                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                     |                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>DESIGN AND IMPLEMENT<br/>MICROCONTROLLER-BASED<br/>PROJECTS</b>                                                                                                                                                                                                                                                                                                                                          | <b>Nominal Duration:</b><br>45 hrs. | <b>Unit Code:</b><br>SICIP-LE-IEE-05-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to design and implement microcontroller-based projects. It specifically includes the tasks of interpreting basic programming for microcontrollers, preparing projects using Arduino UNO and NANO, preparing projects using ATmega 2560, preparing temperature measurement project using STM32F411 and preparing projects using ESP Wroom 32. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

| Elements of Competency                              | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Interpret basic programming for microcontrollers | 1.1 Arduino IDE environment is interpreted and recognized to initiate programming tasks.<br>1.2 Programming language is identified.<br>1.3 <b><u>Microcontrollers</u></b> are interpreted and recognized to the project.<br>1.4 <b><u>Data types</u></b> specific to the microcontroller are identified in programming tasks.<br>1.5 Functions, arrays and pointer within the programming language are defined for specific microcontroller tasks.<br>1.6 Input and output pins of the microcontroller are understood to interact with external devices.                           |
| 2. Prepare projects using Arduino UNO and NANO      | 2.1 Arduino IDE environment is configured and set up for programming tasks.<br>2.2 <b><u>Arduino board</u></b> , and its operations are performed for communication and control.<br>2.3 An LED blinking project is identified, and the code is written and tested on the Arduino UNO or NANO.<br>2.4 <b><u>Digital sensors</u></b> are interfaced with the Arduino board, and their operations are verified.<br>2.5 A Bluetooth/Wi-Fi control circuit is designed, implemented, and operated for remote control of connected devices.<br>2.6 An IR receiver circuit is understood. |
| 3. Prepare projects using ATmega 2560               | 3.1 AC load control with relay module is designed and implemented.<br>3.2 Light dimmer with potentiometer project is set up and its functionality is verified.<br>3.3 Automatic street light control project is created and the system operates based on ambient light conditions.                                                                                                                                                                                                                                                                                                 |
| 4. Prepare temperature                              | 4.1 Temperature sensor is selected and interfaced with the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| measurement project using STM32F411                           | <p>STM32F411 microcontroller.</p> <p>4.2 Sensor's data is collected and processed using appropriate programming techniques.</p> <p>4.3 Program code is written and tested on STM32F411 to ensure proper data acquisition from the sensor.</p> <p>4.4 Temperature data is displayed or logged for further analysis or monitoring.</p> <p>4.5 Project is tested under various temperature conditions.</p> <p>4.6 Final project is compiled, uploaded, and verified for correct functionality.</p>                                                                  |
| 5. Prepare stepper and servo motor control using ESP Wroom 32 | <p>5.1 Stepper motor and servo motor is interfaced with the ESP Wroom 32.</p> <p>5.2 Control circuit is set up for precise motor movement.</p> <p>5.3 Program is written and tested on the ESP Wroom 32 to control the stepper motor's speed and direction.</p> <p>5.4 Program is developed and uploaded to the ESP Wroom 32.</p> <p>5.5 Servo motor and stepper motor control system is verified by testing its ability to rotate.</p> <p>5.6 Final stepper and servo motor control projects are compiled, uploaded, and verified for proper functionality.</p> |

### Range of Variables

| Variable            | Range (Includes but not limited to:)                                                                                                                                                            |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Microcontrollers | <p>1.1 Arduino UNO and NANO</p> <p>1.2 ATmega 2560</p> <p>1.3 STM32F411</p> <p>1.4 ESP Wroom 32</p> <p>1.5 Arduino Mini</p> <p>1.6 Arduino Mega</p> <p>1.7 ESP 8266</p> <p>1.8 Raspberry-PI</p> |
| 2. Data types       | <p>2.1 Boolean</p> <p>2.2 Integer</p> <p>2.3 Float</p> <p>2.4 Character</p> <p>2.5 String</p> <p>2.6 Array</p>                                                                                  |
| 3. Arduino board    | <p>3.1 UNO</p> <p>3.2 NANO</p> <p>3.3 MINI</p> <p>3.4 MEGA</p>                                                                                                                                  |
| 4. Digital sensors  | <p>4.1 Sound sensor</p> <p>4.2 Passive Infrared Sensor (PIR) sensor</p> <p>4.3 Motion sensor</p> <p>4.4 Smoke sensor</p> <p>4.5 Gas sensor</p>                                                  |

## Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | <p>1.1 Identify the Arduino IDE environment and programming language used in microcontroller-based tasks.</p> <p>1.2 Describe the role and structure of microcontrollers in embedded system projects.</p> <p>1.3 Explain the use of data types, functions, arrays, and pointers in microcontroller programming.</p> <p>1.4 Understand the input/output pin configurations for interfacing external components.</p> <p>1.5 Define the process of setting up Arduino and STM32F411 for coding and communication.</p> <p>1.6 Identify the interfacing methods for sensors, actuators, and wireless modules with microcontrollers.</p> <p>1.7 Describe how to implement basic projects like LED blink, IR receiver, and AC load control.</p> <p>1.8 Conceptualize sensor data collection, processing, and display techniques using embedded programming.</p> <p>1.9 Explain the logic behind motor control projects using stepper/servo motors and ESP Wroom 32.</p> |
| 2. Underpinning Skills    | <p>2.1 Configuring the Arduino IDE environment and setting up boards for programming tasks.</p> <p>2.2 Applying microcontroller programming using functions, arrays, pointers, and relevant data types.</p> <p>2.3 Constructing basic projects like LED blink and testing code on Arduino UNO or NANO.</p> <p>2.4 Interfacing digital sensors and verifying their operations with Arduino and STM32F411.</p> <p>2.5 Designing and implementing control circuits like IR receivers, Bluetooth/Wi-Fi modules, and AC relays.</p> <p>2.6 Creating sensor-based automation projects such as light dimmers and street light controllers.</p> <p>2.7 Collecting and processing temperature data using STM32F411 and appropriate programming logic.</p> <p>2.8 Operating stepper and servo motors via ESP Wroom 32 and verifying motion control.</p> <p>2.9 Uploading final programs and testing functionality under real-world conditions.</p>                         |
| 3. Underpinning Attitudes | <p>3.1 Commitment to occupational health and safety</p> <p>3.2 Promptness in carrying out activities</p> <p>3.3 Sincerity and honesty to duties</p> <p>3.4 Environmental concerns</p> <p>3.5 Eagerness to learn</p> <p>3.6 Tidiness and timeliness</p> <p>3.7 Respect for rights of peers and seniors in workplace</p> <p>3.8 Good relationships with peers, sub-ordinates and seniors in workplace</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 4. Resource Implications  | The following resources must be provided:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|  |                                                                                                                                                                               |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>4.1 Workplace (simulated or actual)</p> <p>4.2 Tools, equipment and facilities needed for the process/ activity</p> <p>4.3 Materials relevant to the proposed activity</p> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | <p>Assessment required evidence that the candidate:</p> <p>1.1 interpreted basic programming for microcontrollers</p> <p>1.2 prepared projects using Arduino UNO and NANO.</p> <p>1.3 prepared projects using ATmega 2560.</p> <p>1.4 prepared temperature measurement project using STM32F411</p> <p>1.5 prepared projects using ESP Wroom 32</p> |
| 2. Methods of Assessment          | <p>Competency should be assessed by:</p> <p>2.1 Written test</p> <p>2.2 Practical demonstration</p> <p>2.3 Oral questioning</p> <p>2.4 Portfolio (Optional)</p>                                                                                                                                                                                    |
| 3. Context of Assessment          | <p>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</p> <p>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</p>                                                                           |



|                                                                                                                                                                                                                                                                                                                                                                                              |                                     |                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>PERFORM PCB DESIGN USING ALTIUM SOFTWARE</b>                                                                                                                                                                                                                                                                                                                | <b>Nominal Duration:</b><br>30 hrs. | <b>Unit Code:</b><br>SICIP-LE-IEE-06-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to perform PCB design using ALTIUM software. It specifically includes the tasks of identifying Altium interfaces and tools, carrying out schematic design and diagram for PCB layout, performing auto-manual routing and footprint layout and designing and printing PCB projects using Altium. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

| Elements of Competency                                   | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identify Altium interfaces and tools                  | 1.1 Altium software is defined.<br>1.2 Basic tools of Altium software are identified and their locations within the interface.<br>1.3 Functions and purposes of each tool in Altium are explained.<br>1.4 Design tool options are interpreted.<br>1.5 Interface for creating and editing circuit designs is explored.<br>1.6 Altium's simulation and verification tool options are identified and their roles are understood.                                                                                                                                                     |
| 2. Carry out schematic design and diagram for PCB layout | 2.1 Libraries for Altium are installed and configured to include all necessary components.<br>2.2 Schematic design of a basic electronic circuit is made using appropriate components.<br>2.3 Component connections in the schematic are made, ensuring functionality.<br>2.4 Schematic is reviewed for accuracy, and necessary adjustments are made.<br>2.5 Schematic diagram is converted into a <b><u>PCB layout</u></b> following proper <b><u>conversion steps</u></b> .<br>2.6 Final PCB layout is verified and saved, ensuring that the design meets project requirements. |
| 3. Perform auto-manual routing and footprint layout      | 3.1 Signal trace routing is performed ensuring correct placement.<br>3.2 Power and ground plane routing is performed to ensure a stable electrical system.<br>3.3 High-speed signal routing is carried out considering impedance control and minimizing signal degradation.<br>3.4 Differential plane routing is implemented to maintain signal integrity and reduce noise in differential pairs.<br>3.5 Footprint layout is performed by placing and aligning components based on design specifications.<br>3.6 Routing and layout design is reviewed to ensure                  |

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>compliance with design rules and manufacturability.</p> <p>3.7 Final adjustments are made to the routing and layout to improve performance.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 4. Design and print PCB projects using Altium. | <p>4.1 LED bulb PCB board is designed using Altium, ensuring proper component placement.</p> <p>4.2 Power supply PCB board is designed to power distribution.</p> <p>4.3 Water pump controller PCB is designed for optimal control functionality.</p> <p>4.4 PCB design is verified to ensure compliance with design rules and functionality before printing.</p> <p>4.5 PCB layout is prepared and optimized for the manufacturing process.</p> <p>4.6 Final PCB designs are printed, ensuring accurate component placement.</p> <p>4.7 Design files are exported in the required format for PCB fabrication and verified.</p> |

### Range of Variables

| Variable            | Range (Includes but not limited to:)                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. PCB layout       | <p>1.1 Single layer</p> <p>1.2 Double layer</p> <p>1.3 Multi-layer</p> <p>1.4 Rigid layer</p> <p>1.5 Flexible PCB</p> <p>1.6 Rigid-flex</p> <p>1.7 High frequency</p> <p>1.8 High Density Interconnect (HDI)</p>                                                                                                                                                                                            |
| 2. Conversion steps | <p>2.1 Schematic Capture to PCB Layout</p> <p>2.2 Component Placement</p> <p>2.3 Layer Setup</p> <p>2.4 Signal Routing</p> <p>2.5 Design Rule Check (DRC)</p> <p>2.6 Power and Ground Planes</p> <p>2.7 Via and Hole Placement</p> <p>2.8 Final Design Review</p> <p>2.9 Bill of Materials (BOM) Generation</p> <p>2.10 Gerber File Generation</p> <p>2.11 Assembly Drawings</p> <p>2.12 Final Approval</p> |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                            |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | <p>1.1 Define Altium software and its use in electronic circuit and PCB design.</p> <p>1.2 Identify basic tools in the Altium interface and their respective locations.</p> <p>1.3 Explain the functions and purposes of schematic and</p> |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>PCB design tools.</p> <p>1.4 Interpret design options, simulation features, and verification processes.</p> <p>1.5 Understand how to install libraries and use components in schematic creation.</p> <p>1.6 Describe the steps for converting schematic diagrams into PCB layouts.</p> <p>1.7 Identify routing techniques including signal trace, power plane, and differential routing.</p> <p>1.8 Explain the process of exporting final design files for PCB manufacturing.</p>                                                                                                                                                                                                                                                                                                                                                                                     |
| 2. Underpinning Skills    | <p>2.1 Configuring Altium software tools, libraries, and interface for schematic and PCB design tasks.</p> <p>2.2 Constructing basic schematic diagrams and connecting components with correct functionality.</p> <p>2.3 Converting schematics into PCB layouts and reviewing them for accuracy and completeness.</p> <p>2.4 Performing signal, power, and ground plane routing, including high-speed and differential signal handling.</p> <p>2.5 Designing and placing component footprints as per layout specifications and electrical requirements.</p> <p>2.6 Creating real-world PCB projects like LED bulbs, power supplies, and pump controllers.</p> <p>2.7 Verifying design functionality through simulation and optimizing layout for performance.</p> <p>2.8 Exporting design files and preparing final PCBs for fabrication with proper file formatting.</p> |
| 3. Underpinning Attitudes | <p>3.1 Commitment to occupational health and safety</p> <p>3.2 Promptness in carrying out activities</p> <p>3.3 Sincerity and honesty to duties</p> <p>3.4 Environmental concerns</p> <p>3.5 Eagerness to learn</p> <p>3.6 Tidiness and timeliness</p> <p>3.7 Respect for rights of peers and seniors in workplace</p> <p>3.8 Good relationships with peers, sub-ordinates and seniors in workplace</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4. Resource Implications  | <p>The following resources must be provided:</p> <p>4.1 Workplace (simulated or actual)</p> <p>4.2 Tools, equipment and facilities needed for the process/ activity</p> <p>4.3 Materials relevant to the proposed activity</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | <p>Assessment required evidence that the candidate:</p> <ul style="list-style-type: none"> <li>1.1 identified Altium interfaces and tools</li> <li>1.2 carried out schematic design and diagram for PCB layout</li> <li>1.3 performed auto-manual routing and footprint layout</li> <li>1.4 designed and printed PCB projects using Altium.</li> </ul> |
| 2. Methods of Assessment          | <p>Competency should be assessed by:</p> <ul style="list-style-type: none"> <li>2.1 Written test</li> <li>2.2 Practical demonstration</li> <li>2.3 Oral questioning</li> <li>2.4 Portfolio (Optional)</li> </ul>                                                                                                                                       |
| 3. Context of Assessment          | <ul style="list-style-type: none"> <li>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</li> <li>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</li> </ul>                                  |

|                                                                                                                                                                                                                                                                                                                                                                               |                                     |                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|
| <b>Unit of Competency:</b><br><b>IMPLEMENT SOLAR AND WIND ENERGY SYSTEMS USING HYBRID CONTROLLERS</b>                                                                                                                                                                                                                                                                         | <b>Nominal Duration:</b><br>15 hrs. | <b>Unit Code:</b><br>SICIP-LE-IEE-07-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to implement solar and wind energy systems using hybrid controllers. It specifically includes the tasks of interpreting solar energy systems, designing solar home lighting system, interpreting wind energy systems and operating solar & wind energy using hybrid controllers. |                                     |                                        |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

| Elements of Competency               | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Interpret solar energy systems    | 1.1 <b><u>Solar energy system</u></b> components are identified and their functions are understood.<br>1.2 Basic principles of photovoltaic (PV) energy conversion are interpreted.<br>1.3 System capacity requirements are identified based on energy needs.<br>1.4 Electrical connections and configurations for solar energy systems are interpreted.<br>1.5 Solar energy system diagrams and schematics are identified.<br>1.6 Performance of a solar energy system is understood.                                                                                                                                                                                       |
| 2. Design solar home lighting system | 2.1 Solar energy system requirements are identified based on the lighting needs of the home.<br>2.2 Solar panel specifications are determined to meet the energy demands of the lighting system.<br>2.3 Battery capacity and type are selected based on the system's energy storage requirements.<br>2.4 Charge controller is selected and configured to regulate the charging and discharging of the battery.<br>2.5 Wiring and connections are designed between the solar panel, battery, and lighting units.<br>2.6 Design is reviewed to ensure proper functionality of the home lighting system.<br>2.7 System design is verified for compliance with safety standards. |
| 3. Interpret wind energy systems     | 3.1 Basic principles of wind energy conversion are understood.<br>3.2 <b><u>Key components of a wind energy system</u></b> are identified and their functions are understood.<br>3.3 Energy requirements for a wind energy system are interpreted.<br>3.4 Wind energy system diagrams and schematics are                                                                                                                                                                                                                                                                                                                                                                     |

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                         | identified.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                         | 3.5 System efficiency is interpreted based on factors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 4. Operate solar & wind energy using hybrid controllers | 4.1 Hybrid controller is set up and configured to manage both solar and wind energy inputs.<br>4.2 Solar and wind energy systems are connected to the hybrid controller.<br>4.3 Hybrid controller's settings are adjusted to optimize energy storage.<br>4.4 System performance is monitored to ensure the hybrid controller operates within the required parameters.<br>4.5 Controller's energy distribution function is verified.<br>4.6 Data from the hybrid controller is analyzed to assess the efficiency and performance.<br>4.7 Troubleshooting and corrective actions are performed on the hybrid controller. |

### Range of Variables

| Variable                                  | Range (Includes but not limited to:)                                                                                                                                                   |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Solar energy system                    | 1.1 Off-grid solar system<br>1.2 On-grid solar system                                                                                                                                  |
| 2. Key components of a wind energy system | 2.1 Wind turbine<br>2.2 Hybrid charge controller<br>2.3 Tower<br>2.4 Yaw Mechanism<br>2.5 Anemometer and Wind Vane<br>2.6 Electrical Grid Connection<br>2.7 Battery Storage (Optional) |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Identify the components of solar and wind energy systems and explain their functions.<br>1.2 Understand the basic principles of photovoltaic (PV) and wind energy conversion.<br>1.3 Define system capacity based on energy consumption and lighting requirements.<br>1.4 Describe electrical wiring, configurations, and schematic diagrams for renewable systems.<br>1.5 Explain the roles of batteries, charge controllers, and hybrid controllers in energy storage and distribution.<br>1.6 Interpret solar panel and battery specifications to match design needs.<br>1.7 Understand how to monitor performance and ensure safety compliance in energy systems.<br>1.8 Explain efficiency analysis and troubleshooting methods for hybrid renewable systems. |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2. Underpinning Skills    | 2.1 Configuring components of solar and wind energy systems for residential use.<br>2.2 Designing system wiring between solar panels, batteries, charge controllers, and lighting units.<br>2.3 Constructing and reviewing solar energy system diagrams and verifying functionality and safety.<br>2.4 Applying basic PV and wind energy principles to configure efficient renewable systems.<br>2.5 Adjusting hybrid controller settings and monitoring energy storage and system performance.<br>2.6 Analyzing controller data to assess efficiency and performing necessary troubleshooting.<br>2.7 Implementing safe and optimized hybrid energy solutions for reliable power distribution. |
| 3. Underpinning Attitudes | 1.1 Commitment to occupational health and safety<br>1.2 Promptness in carrying out activities<br>1.3 Sincerity and honesty to duties<br>1.4 Environmental concerns<br>1.5 Eagerness to learn<br>1.6 Tidiness and timeliness<br>1.7 Respect for rights of peers and seniors in workplace<br>1.8 Good relationships with peers, sub-ordinates and seniors in workplace                                                                                                                                                                                                                                                                                                                            |
| 4. Resource Implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)<br>4.2 Tools, equipment and facilities needed for the process/activity<br>4.3 Materials relevant to the proposed activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 interpreted solar energy systems<br>1.2 designed solar home lighting system<br>1.3 interpreted wind energy systems<br>1.4 operated solar & wind energy using hybrid controllers |
| 2. Methods of Assessment          | Competency should be assessed by:<br>2.1 Written test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                                                                                                |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.                                                                                       |

|  |                                                                                                          |
|--|----------------------------------------------------------------------------------------------------------|
|  | 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert. |
|--|----------------------------------------------------------------------------------------------------------|



### **Workshop/Lab Facility Standard**

|                            |                                                   |
|----------------------------|---------------------------------------------------|
| <b>Course Name:</b>        | Industrial Electrical and Electronics Application |
| <b>Number of Trainees:</b> | 25                                                |

#### **Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):**

- Classroom – 350 sft
- Workshop/ lab – 1000 sft

#### **Major Training Equipment and Training Facilities:**

| <b>Sl. No.</b> | <b>Major Equipment and Training facilities</b>                           | <b>Required facilities</b> |
|----------------|--------------------------------------------------------------------------|----------------------------|
| 1.             | Computer/Laptop with internet facility                                   | 01                         |
| 2.             | Multimedia projector & screen                                            | 01                         |
| 3.             | Printer                                                                  | 01                         |
| 4.             | Internet connectivity                                                    | 01                         |
| 5.             | Resistors, capacitors, inductors, switches, breadboards and jumper wires | 350                        |
| 6.             | Multimeter for measuring voltage, current, and resistance                | 20                         |
| 7.             | DC Power Supply                                                          | 10                         |
| 8.             | Oscilloscope for signal analysis                                         | 10                         |
| 9.             | Function Generator for waveform generation                               | 10                         |
| 10.            | Diodes, transistors, operational amplifiers (Op-Amps)                    | 150                        |
| 11.            | Logic gates (AND, OR, NOT), BJT, MOSFET, SCR, TRIAC, DIAC                | 200                        |
| 12.            | Relays (electromechanical and solid-state)                               | 30                         |
| 13.            | Timers and counters                                                      | 60                         |
| 14.            | AC and DC motor control kits                                             | 10                         |
| 15.            | Variable Frequency Drive (VFD)                                           | 12                         |
| 16.            | Contactors and overload relays                                           | 50                         |
| 17.            | PLC hardware (Siemens)                                                   | 10                         |
| 18.            | Ladder diagram software (TIA Portal)                                     | 10                         |
| 19.            | PLC programming cables                                                   | 10                         |
| 20.            | Proximity sensors and photo sensors                                      | 40                         |
| 21.            | Push buttons, switches, and indicator lights                             | 200                        |
| 22.            | Communication interface for PLC-PC integration                           | 10                         |
| 23.            | Arduino UNO and NANO kits                                                | 20                         |
| 24.            | ATmega 2560 development board                                            | 10                         |
| 25.            | STM32F411 development kit                                                | 10                         |
| 26.            | ESP Wroom 32 development board                                           | 10                         |
| 27.            | LCD displays                                                             | 10                         |
| 28.            | Servo motors, DC motors, and stepper motor                               | 30                         |

|     |                                                             |    |
|-----|-------------------------------------------------------------|----|
| 29. | Computers with Altium Designer software installed           | 10 |
| 30. | PCB fabrication kit (copper-clad boards, etching solutions) | 40 |
| 31. | Soldering stations and tools                                | 10 |
| 32. | PCB stencil and screen-printing tools                       | 10 |
| 33. | PCB test equipment (continuity tester)                      | 10 |
| 34. | Solar panels (various wattage ratings)                      | 10 |
| 35. | Hybrid controller systems for solar and wind energy         | 10 |
| 36. | Wind turbine kits (small-scale models)                      | 10 |
| 37. | Charge controllers                                          | 10 |
| 38. | Inverters for DC to AC conversion                           | 10 |
| 39. | Batteries for energy storage                                | 10 |
| 40. | Analog sensors, 0-10V                                       | 10 |
| 41. | Servo motor with driver                                     | 02 |

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Industrial Electrical and Electronics Application, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.